Analyzing Retail Store Sales

Introduction

A large retail store chain wants to analyze its sales data to identify trends, improve inventory management, and boost revenue. The dataset includes sales records for multiple stores over a period, capturing details like product categories, sales revenue, and customer demographics. Your task is to analyze this data using NumPy and Pandas, and visualize key insights using Matplotlib.

Scenario & Dataset

The dataset contains the following columns:

- **Store_ID**: Unique identifier for each store
- **Product_Category**: Category of the product sold
- **Units_Sold**: Number of units sold
- **Price_Per_Unit**: Price of a single unit of the product
- **Total_Sales**: Revenue generated from the product (Units_Sold * Price_Per_Unit)
- **Date**: Date of the sale
- **Customer_Age**: Age of the customer purchasing the product
- **Customer_Gender**: Gender of the customer
- **Store_Location**: City where the store is located

You are required to perform the following analyses.

DATASET: LINK

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NumPy Operations

Task 1: Basic Data Exploration

1. Load the dataset and convert it into a NumPy array.
2. Find the total revenue generated across all stores.
3. Identify the highest and lowest price per unit among all products.
4. Calculate the average number of units sold per product.

Task 2: Store Performance Analysis

1. Determine the store with the highest total revenue.
2. Identify the store with the lowest number of units sold.
3. Compute the standard deviation of total sales across all stores.

Pandas Operations

Task 3: Data Aggregation

1. Group the data by Product_Category and compute the total revenue for each category.
2. Find the top 3 best-selling product categories in terms of revenue.
3. Calculate the average price per unit for each product category.

Task 4: Customer Analysis

1. Analyze the proportion of sales generated by male vs. female customers.
2. Determine the store location that has the highest number of unique customers.

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Matplotlib Visualizations

Task 5: Sales Trends Over Time

1. Plot a line chart showing total sales per month over the available period.
2. Create a bar chart to compare total revenue among different store locations.
3. Generate a pie chart to display the revenue share of different product categories.

Task 6: Customer Demographics Visualization

1. Create a histogram to visualize the age distribution of customers.
2. Plot a gender-wise sales comparison using a bar chart.

Bonus Questions

Task 7: Advanced Analysis

1. Identify whether there is any seasonality in sales data (e.g., peaks in certain months).
2. Find the correlation between customer age and total sales.
3. Predict the expected sales for the next month using the average monthly sales trend.

Conclusion

By completing this case study, you will gain hands-on experience in using NumPy for numerical computations, Pandas for data manipulation, and Matplotlib for data visualization. These insights can help businesses make data-driven decisions to optimize their operations and maximize revenue.